



FULL PROFILE • REAL-DATA RUN (LIVE MCP)

Challenge — Notch It Up

Memory-triage forensic report • single Windows x64 RAM image • host notch-it-up

REPORT ID	dc58027639681018eee8bc948b5d8c72b06dc2bdc6405c0064ee91e154b9526a
CASE ID	CHALLENGE-NOTCHITUP
CASE NAME	Challenge - Notch It Up
EXAMINER	victor.galvan
PROFILE	full
INCIDENT TYPE	dfir
SEVERITY	medium (case) • high (top finding)
SNAPSHOT	2026-06-07T03:37:21.141686+00:00
GENERATED	2026-06-07T03:41:32Z
MCP HOST	

SEALED • report_id dc58027639681018... • 5 approved findings • top seal hmac-sha256:bbbe885b29ff07418442af... • examiner victor.galvan

Executive Summary

A single 1.6 GB raw memory image (Challenge.raw, SHA-256 80366d7e...1407b23) from the VirtualBox Windows x64 guest **notch-it-up** (boot 2019-08-19 14:40:07 UTC, 53 processes) was triaged end-to-end over the live Agentropix-SIFT MCP. Volatility3 malfind surfaced four PAGE_EXECUTE_READWRITE (RWX) regions across four host processes; the standout is a 64 KB RWX VAD in **explorer.exe (PID 1944)** carrying executable indirect-jump shellcode bytes — the classic code-injection signature (MITRE T1055). A 512 KB RWX region in **WmiPrvSE.exe (PID 2292)** adds a WMI-execution dimension (T1047). Five findings were recorded and sealed by the report engine (2 high, 2 medium, 1 info). Scope is RAM-only: there is no on-disk or external-infrastructure visibility, so the initial-access vector and dwell time are not determinable from this evidence.

Headline: Memory-resident code injection confirmed in explorer.exe (PID 1944) — a 64 KB RWX VAD at 0x4320000 holding shellcode (41 ba 80 00 00 00 48 b8 ... 48 ff 20, payload SHA-256 65196e1a...1726ca6f), plus a large 512 KB RWX region in WmiPrvSE.exe (PID 2292). T1055 Process Injection confirmed; T1047 WMI execution suspected. **Recommended:** isolate the host and acquire the disk image to recover the injector and persistence — RAM alone cannot establish the entry vector.

5

APPROVED FINDINGS

2 high, 2 medium, 1 info

1

HOSTS IN SCOPE

notch-it-up (single RAM image)

5

IOCS CATALOGUED

2 memory_region, 2 process, 1 sha256

N/D

DWELL / WINDOW

single RAM snapshot — no dwell window

2

ATT&CK TECHNIQUES

T1047, T1055

Inconcl.

INITIAL ACCESS

vector not in RAM scope

Hosts / evidence roster



notch-it-up

Windows x64 guest ·
compromised · 4 RWX injection hits



Challenge.raw

1.6 GB raw RAM image · sole evidence



10.0.2.15

VirtualBox NAT guest IP ·
browser sessions only

02 · RISK ASSESSMENT

Risk Matrix

Two findings score in the high band: the explorer.exe shellcode injection (impact 5 / likelihood 5 — confirmed executable bytes, high confidence) and the 512 KB WmiPrvSE.exe RWX region (impact 5 / likelihood 4). The remaining zeroed/small RWX regions and the chain-of-custody info record sit in the medium-to-low band. *Scoring is qualitative (operator-derived from approved-finding severity + technique), not a tool-computed number.*

5

F4

F1

4

3

2

1

1

2

3

4

5

F2

F3

F5

X-axis: Likelihood (1-5) · Y-axis: Impact (1-5) · Risk = Impact × Likelihood

REF	RISK	IMPACT	LIKELIHOOD	SCORE	SEVERITY
F-NOTCH-001	Code injection in explorer.exe (shellcode)	5	5	25	HIGH
F-NOTCH-004	Large RWX region in WmiPrvSE.exe (WMI host)	5	4	20	HIGH
F-NOTCH-002	Second zeroed RWX VAD in explorer.exe	3	3	9	MEDIUM
F-NOTCH-003	RWX region in chrome.exe	2	3	6	MEDIUM
F-NOTCH-005	Evidence chain-of-custody record	1	1	1	INFO

03 · KEY FINDINGS

Key Findings

HIGH

F-NOTCH-001

PAGE_EXECUTE_READWRITE injected region in explorer.exe (PID 1944)

malfind flagged a 64KB PAGE_EXECUTE_READWRITE VAD at 0x4320000 in explorer.exe (PID 1944) carrying executable indirect-jump shellcode bytes (41 ba 80 00 00 00 48 b8 ... 48 ff 20). Classic code-injection signature.

MEDIUM

F-NOTCH-002

PAGE_EXECUTE_READWRITE zeroed RWX region in explorer.exe (PID 1944)

malfind flagged a second 4KB PAGE_EXECUTE_READWRITE VAD at 0x3ce0000 in explorer.exe (PID 1944), zeroed - a second RWX VAD in the same injected host process.

MEDIUM

F-NOTCH-003

PAGE_EXECUTE_READWRITE RWX region in chrome.exe (PID 2124)

malfind flagged a 4KB PAGE_EXECUTE_READWRITE VAD at 0x4830000 in chrome.exe (PID 2124), zeroed.

HIGH

F-NOTCH-004

Large PAGE_EXECUTE_READWRITE region in WmiPrvSE.exe (PID 2292)

malfind flagged a 512KB PAGE_EXECUTE_READWRITE VAD at 0x1bd0000 in WmiPrvSE.exe (PID 2292) - WMI provider host carrying a large RWX region.

INFO

F-NOTCH-005

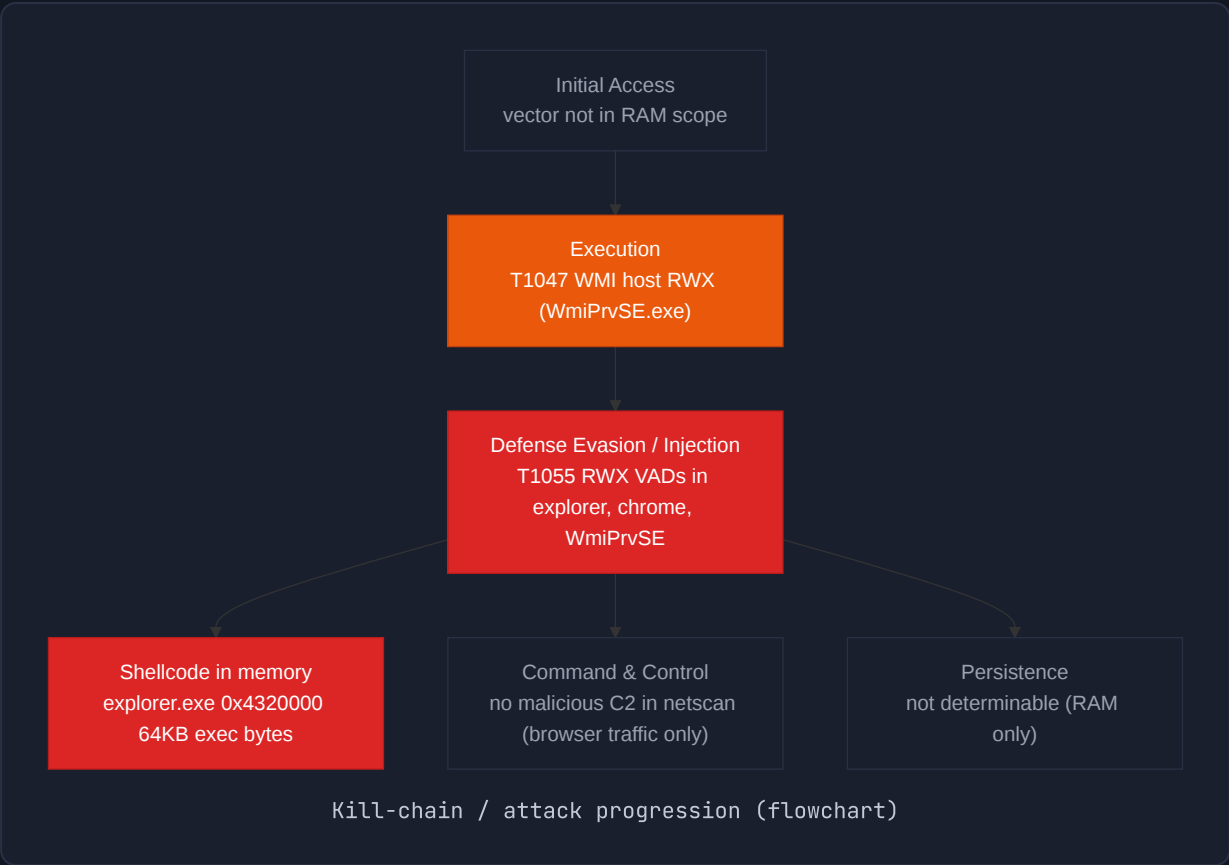
Evidence image registered (chain-of-custody hash)

evidence_register computed the chain-of-custody SHA-256 for the 1.6GB raw memory image.

Attack Chain & MITRE ATT&CK

Reconstructed from the RAM snapshot alone, the observable adversary activity is post-compromise memory-resident code execution. Volatility3 confirms RWX regions injected into legitimate host processes (T1055 Process Injection), with a WMI-provider host (WmiPrvSE.exe) carrying a large RWX region consistent with WMI-based execution (T1047). The initial-access, command-and-control and persistence phases

are *not reconstructable* from a single memory image — the kill-chain below shows only the phases this evidence substantiates; greyed nodes are inconclusive.



MITRE ATT&CK techniques observed

DEFENSE EVASION / PRIVILEGE ESCALATION T1055 Process Injection Confirmed: RWX (PAGE_EXECUTE_READWRITE) VADs injected into explorer.exe (PID 1944, 64KB exec shellcode + 4KB zeroed), chrome.exe (PID 2124) and WmiPrvSE.exe (PID 2292). 4 malfind hits total.	EXECUTION T1047 Windows Management Instrumentation Suspected: a 512KB RWX region in the WMI provider host WmiPrvSE.exe (PID 2292) is consistent with WMI-based code execution. Mapped on finding F-NOTCH-004.
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IOC Catalogue

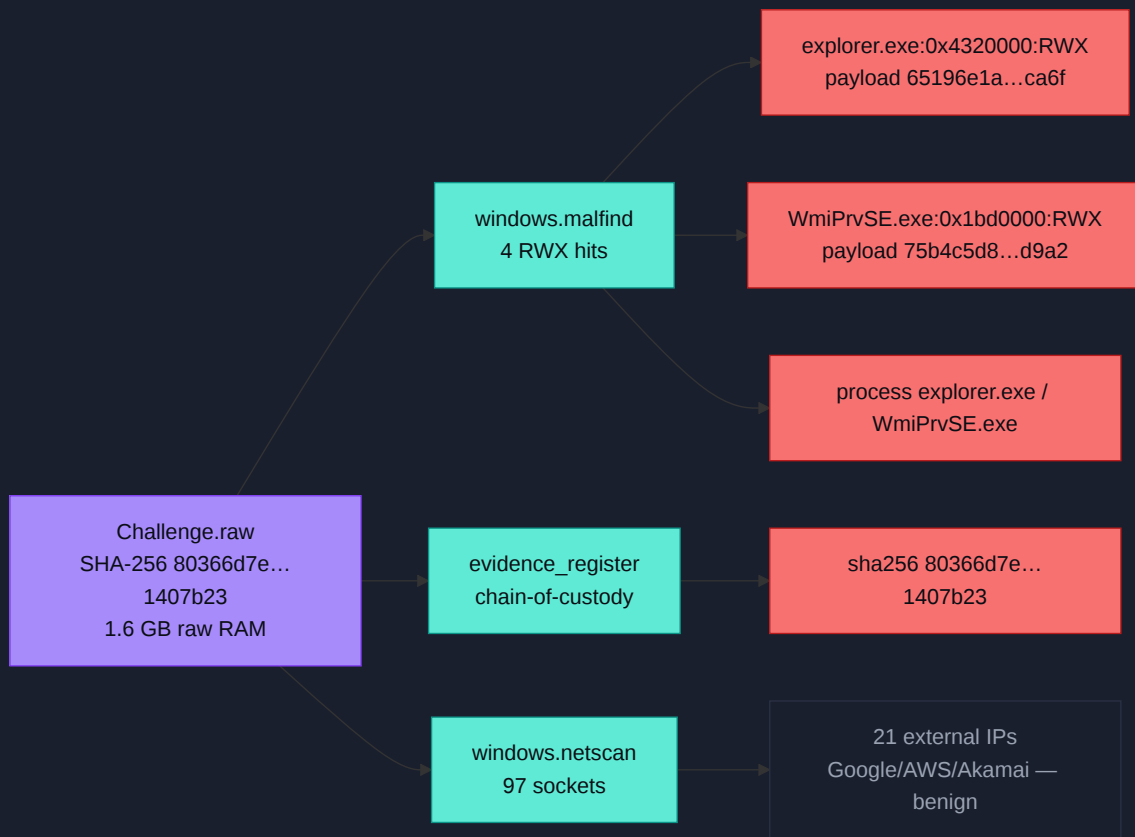
5 indicators were catalogued from the memory image: 2 injected RWX memory-region IOCs, 2 abused host-process IOCs, and 1 evidence chain-of-custody SHA-256. All are **evidence-internal** (recovered from inside the RAM image) — no external network IOC

was confirmed malicious; the netscan traffic resolved to ordinary Google/AWS browser endpoints.

TYPE	VALUE	ROLE	CONFIDENCE	MITRE	SOURCE
memory_region	explorer.exe:PID1944:0x4320000:RWX	Injected RWX region (shellcode)	High	T1055	windows.malfind / F-NOTCH-001
memory_region	WmiPrvSE.exe:PID2292:0x1bd0000:RWX	Injected RWX region (512KB)	High	T1047, T1055	windows.malfind / F-NOTCH-004
process	explorer.exe	Injected host process	High	T1055	windows.malfind / F-NOTCH-001
process	WmiPrvSE.exe	Injected host process (WMI)	High	T1047, T1055	windows.malfind / F-NOTCH-004
sha256	80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23	Evidence image hash (chain-of-custody)	N/A	—	evidence_register / F-NOTCH-005

Negative space (surveyed, not confirmed malicious): external IPv4 endpoints — 21 distinct ESTABLISHED foreign addresses were recovered (Google 172.217/216.58 ranges, AWS 52.24.89.101 / 54.149.112.164, Akamai 23.195.74.19), all consistent with normal Firefox/Chrome browsing from guest 10.0.2.15 — none was classified as C2. Malicious file hashes, suspicious domains, and rogue accounts: none surfaced in this image (no on-disk artefacts available to test against).

IOC provenance



IOC → evidence-source provenance graph

Host Artefacts

Single Windows x64 memory image — there are no on-disk artefacts to corroborate. Volatility3 matched the kernel symbol table and recovered all 53 processes (boot 2019-08-19 14:40:07 UTC). The host is a VirtualBox guest (VBoxService/VBoxTray present). The process table below highlights the four injected hosts (red rows from malfind) and notable context processes: DumpIt.exe (PID 4084) — the memory acquisition tool itself, run by user Jaffa — and WinRAR.exe (PID 3716) which opened C:\Users\Jaffa\Desktop\pr0t3ct3d\flag.rar (the CTF objective). No svcsan was captured in this run, so the services table is not applicable; RWX VAD evidence is memory-only.

notch-it-up — Windows x64 (53 processes)

PID	PPID	IMAGE	THR	STARTED (UTC)	COMMAND LINE	NOTES
4	0	System	78	2019-08-19 14:40:07	—	
264	4	smss.exe	2	2019-08-19 14:40:07	—	
336	328	csrss.exe	10	2019-08-19 14:40:10	—	
384	328	wininit.exe	3	2019-08-19 14:40:11	—	
396	376	csrss.exe	9	2019-08-19 14:40:11	—	
436	376	winlogon.exe	6	2019-08-19 14:40:11	—	
480	384	services.exe	9	2019-08-19 14:40:11	—	
496	384	lsass.exe	7	2019-08-19 14:40:11	—	
504	384	lsmd.exe	10	2019-08-19 14:40:11	—	
608	480	svchost.exe	10	2019-08-19 14:40:11	—	
668	480	VBBoxService.exe	13	2019-08-19 14:40:11	VBBoxService.exe	VirtualBox guest service
724	480	svchost.exe	6	2019-08-19 14:40:11	—	
780	480	svchost.exe	19	2019-08-19 14:40:11	—	
896	480	svchost.exe	22	2019-08-19 14:40:11	—	

PID	PPID	IMAGE	THR	STARTED (UTC)	COMMAND LINE	NOTES
				2		
948	480	svchost.exe	35	2019-08-19 14:40:12	—	
1008	780	audiodg.exe	7	2019-08-19 14:40:12	—	
400	480	svchost.exe	13	2019-08-19 14:40:12	—	
1052	480	svchost.exe	17	2019-08-19 14:40:12	—	
1176	480	spoolsv.exe	14	2019-08-19 14:40:13	—	
1212	480	svchost.exe	21	2019-08-19 14:40:13	—	
1308	480	svchost.exe	17	2019-08-19 14:40:13	—	
1812	480	taskhost.exe	9	2019-08-19 14:40:18	—	
1868	896	dwm.exe	4	2019-08-19 14:40:18	—	
1876	948	taskeng.exe	5	2019-08-19 14:40:18	—	
1944	1844	explorer.exe	35	2019-08-19 14:40:19	C:\Windows\Explorer.EXE	△ 2 RWX VADs (injected host)
1292	1928	GoogleCrashHan	7	2019-08-19 14:40:19	—	
924	1928	GoogleCrashHan	6	2019-08-19 14:40:19	—	

PID	PPID	IMAGE	THR	STARTED (UTC)	COMMAND LINE	NOTES
1108	1944	VBoxTray.exe	14	2019-08-19 14:40:20	"VBoxTray.exe"	VirtualBox guest
880	1944	cmd.exe	1	2019-08-19 14:40:26	"C:\Windows\system32\cmd.exe"	
916	396	conhost.exe	3	2019-08-19 14:40:26	—	
856	480	SearchIndexer.exe	13	2019-08-19 14:40:27	—	
2124	1944	chrome.exe	27	2019-08-19 14:40:46	"...\chrome.exe"	⚠ RWX VAD 0x4830000
2132	2124	chrome.exe	9	2019-08-19 14:40:46	—	
2168	2124	chrome.exe	3	2019-08-19 14:40:49	—	
2292	608	WmiPrvSE.exe	13	2019-08-19 14:40:52	C:\Windows\system32\wbem\wmioprse.exe	⚠ 512KB RWX VAD 0x1bd0000 (WMI)
2340	2124	chrome.exe	12	2019-08-19 14:40:52	—	
2440	2124	chrome.exe	13	2019-08-19 14:40:54	—	
2452	2124	chrome.exe	14	2019-08-19 14:40:54	—	
2800	480	WmiApSrv.exe	6	2019-08-19 14:40:57	—	
2896	608	WmiPrvSE.exe	7	2019-08-19 14:40:57	—	

PID	PPID	IMAGE	THR	STARTED (UTC)	COMMAND LINE	NOTES
2940	2124	chrome.exe	9	2019-08-19 14:41:06	—	
2080	3060	firefox.exe	59	2019-08-19 14:41:08	"...\firefox.exe"	
2860	2080	firefox.exe	11	2019-08-19 14:41:09	—	
3016	2080	firefox.exe	31	2019-08-19 14:41:10	—	
2968	2080	firefox.exe	22	2019-08-19 14:41:11	—	
3316	2080	firefox.exe	21	2019-08-19 14:41:13	—	
3716	1944	WinRAR.exe	7	2019-08-19 14:41:43	"...\WinRAR.exe" "C:\Users\Jaffa\Desktop\pr0t3ct3d\flag.rar"	Opened flag.rar (CTF artefact)
4084	1944	DumpIt.exe	5	2019-08-19 14:41:55	"C:\Users\Jaffa\Desktop\DumpIt.exe"	Memory acquisition tool (DumpIt)
4092	396	conhost.exe	2	2019-08-19 14:41:55	—	
1224	480	sppsvc.exe	5	2019-08-19 14:42:39	—	
2256	2396	GoogleUpdate.exe	3	2019-08-19 14:42:40	—	
1192	2256	GoogleCrashHan	3	2019-08-19 14:42:41	—	

PID	PPID	IMAGE	THR	STARTED (UTC)	COMMAND LINE	NOTES
864	2256	GoogleCra shHan	1	2019-08-1 9 14:42:4 1	—	

Services table (svcsan) — **not applicable**: svcsan was not part of this triage run. The 4 RWX regions are memory-resident artefacts with no on-disk service backing observed.

Process tree



Network Artefacts

windows.netscan recovered 97 sockets. Established external sessions belong to the browsers (firefox.exe PID 2080, chrome.exe PID 2124) reaching 21 distinct Google / AWS / Akamai IPs — ordinary web traffic, not C2. The listeners are stock Windows services (SMB 445/139, RPC 135, EPM 49152-49156). No injected process (explorer 1944 / WmiPrvSE 2292) held a network socket — the injection is memory-resident with no observable beacon in this snapshot. All addresses are evidence-internal (from inside the RAM image).

PROCESS (PID)	LOCAL	REMOTE	STATE	PURPOSE	SOURCE
firefox.exe (2080)	10.0.2.15:4 9235	172.217.19 4,189:443	ESTABLISHED	HTTPS to Google range — browsing	windows.netscan

PROCESS (PID)	LOCAL	REMOTE	STATE	PURPOSE	SOURCE
firefox.exe (2080)	10.0.2.15:49232	172.217.160.131:80	ESTABLISHED	HTTP to Google range — browsing	windows.netscan
firefox.exe (2080)	10.0.2.15:49198	216.58.197.67:443	ESTABLISHED	HTTPS to Google range — browsing	windows.netscan
firefox.exe (2968)	127.0.0.1:49171	127.0.0.1:49170	ESTABLISHED	Loopback IPC (browser internal)	windows.netscan
chrome.exe (2124)	0.0.0.0:5353	*:0 (UDP)	—	mDNS multicast (normal)	windows.netscan
System (4)	0.0.0.0:445	0.0.0.0 (LISTEN)	LISTENING	SMB listener (stock Windows)	windows.netscan
svchost.exe (724)	0.0.0.0:135	0.0.0.0 (LISTEN)	LISTENING	RPC endpoint mapper (stock)	windows.netscan
lsass.exe (496)	0.0.0.0:49156	0.0.0.0 (LISTEN)	LISTENING	RPC (stock Windows)	windows.netscan

No beacon / pivot observed. The injected processes carry no network sockets in this snapshot, and none of the 21 external endpoints resolves to known-malicious infrastructure (all Google/AWS/Akamai browser traffic). Absence of a C2 channel in RAM does not exclude one — it may post-date the snapshot or use a covert channel; disk + flow data would be needed to confirm.

Detailed Findings & Evidence

HIGH F-NOTCH-001

PAGE_EXECUTE_READWRITE injected region in explorer.exe (PID 1944)

malfind flagged a 64KB PAGE_EXECUTE_READWRITE VAD at 0x4320000 in explorer.exe (PID 1944) carrying executable indirect-jump shellcode bytes (41 ba 80 00 00 00 48 b8 ...

48 ff 20). Classic code-injection signature.

EVIDENCE

windows.malfind: PID 1944 explorer.exe VAD 0x4320000 PAGE_EXECUTE_READWRITE 65536B (executable bytes 41 ba 80 00 00 00 48 b8 ...)

carved region payload_sha256 65196e1a65d8e4bfcf42f03b7db79cd07a2573f57c6aad40a97c37791726ca6f

source_evidence_sha256 80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23

MITRE T1055 · CONFIDENCE 0.9 · SEAL hmac-sha256:bbbe885b29ff07418442af3fdf784e3e366cbea20ce5459617608a50ed394500 · RUN 3dee10c2

MEDIUM

F-NOTCH-002

PAGE_EXECUTE_READWRITE zeroed RWX region in explorer.exe (PID 1944)

malfind flagged a second 4KB PAGE_EXECUTE_READWRITE VAD at 0x3ce0000 in explorer.exe (PID 1944), zeroed - a second RWX VAD in the same injected host process.

EVIDENCE

windows.malfind: PID 1944 explorer.exe VAD 0x3ce0000 PAGE_EXECUTE_READWRITE 4096B (zeroed)

carved region payload_sha256 3da12179ac97a8fdcfbdfc5318ba0da46974e1c1ae648dc37e861a926e8563ca

source_evidence_sha256 80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23

MITRE T1055 · CONFIDENCE 0.6 · SEAL hmac-sha256:6c1749e9807c2ede560d8151fd7ad96341dc29372322d2fb024841cdcf09f30 · RUN 87675539

MEDIUM

F-NOTCH-003

PAGE_EXECUTE_READWRITE RWX region in chrome.exe (PID 2124)

malfind flagged a 4KB PAGE_EXECUTE_READWRITE VAD at 0x4830000 in chrome.exe (PID 2124), zeroed.

EVIDENCE

windows.malfind: PID 2124 chrome.exe VAD 0x4830000 PAGE_EXECUTE_READWRITE 4096B (zeroed)

carved region payload_sha256 3243bcc13c7c564288355046c5bc1123888837aeb8ada9ed5cf7800d9c462c1d

source_evidence_sha256 80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23

MITRE T1055 · CONFIDENCE 0.6 · SEAL hmac-sha256:16c7a4ea40d791634a3f2fda35fb8760343d85d412bc790000e0c3818f5eb4db · RUN 5d9034d7

HIGH

F-NOTCH-004

Large PAGE_EXECUTE_READWRITE region in WmiPrvSE.exe (PID 2292)

malfind flagged a 512KB PAGE_EXECUTE_READWRITE VAD at 0x1bd0000 in WmiPrvSE.exe (PID 2292) - WMI provider host carrying a large RWX region.

EVIDENCE

windows.malfind: PID 2292 WmiPrvSE.exe VAD 0x1bd0000 PAGE_EXECUTE_READWRITE 524288B (512KB)
carved_region payload_sha256 75b4c5d8fc85525f97106ddf84197731942bb9acb2d992de50b8a4b7bcfad9a2
source_evidence_sha256 80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23
MITRE T1047, T1055 · CONFIDENCE 0.9 · SEAL hmac-sha256:1907771e425d4e0b381a2571638d4f2cbf201dcb4086d51c68f2444b033c69c6 · RUN 09ea803f

INFO

F-NOTCH-005 Evidence image registered (chain-of-custody hash)

evidence_register computed the chain-of-custody SHA-256 for the 1.6GB raw memory image.

EVIDENCE

evidence_register: /cases/Challenge_NotchItUp/Challenge.raw sha256 80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23 size 1610547200 bytes
source_evidence_sha256 80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23
MITRE — · CONFIDENCE 0.9 · SEAL hmac-sha256:24963789b016f28fd1bad0b10db3cf9206bb13dfaf86333493940540401f5b0 · RUN c1d05f9d

Timeline (UTC)

TIME (UTC)	HOST	EVENT	TECHNIQUE	SOURCE
2019-08-19T14:40:07Z	notch-it-up	System boot - System (PID 4) and smss.exe (PID 264) started; Windows x64 kernel symbols matched by Volatility3 (53 processes)	—	windows.pslist
2019-08-19T14:40:11Z	notch-it-up	Core Windows service stack started: wininit.exe	—	windows.pslist

TIME (UTC)	HOST	EVENT	TECHNIQUE	SOURCE
		(384), services.exe (480), lsass.exe (496), svchost.exe (608), VBoxService.exe (668) - VirtualBox guest		
2019-08-19T14:40:07Z	notch-it-up	Code injection observed: explorer.exe (PID 1944) holds RWX VAD at 0x4320000 with 64KB executable shellcode (41 ba ... 48 b8 ...) and a zeroed RWX VAD at 0x3ce0000	T1055	windows.malfind
2019-08-19T14:40:07Z	notch-it-up	WmiPrvSE.exe (PID 2292) carries a 512KB RWX VAD at 0x1bd0000; chrome.exe (PID 2124) holds a 4KB RWX VAD at 0x4830000	T1047/T1055	windows.malfind
2019-08-19T14:40:11Z	notch-it-up	Active browser network sessions in memory: firefox.exe (PID 2080) ESTABLISHED to Google ranges (172.217.160.131:80, 172.217.194.189:443) from guest 10.0.2.15; chrome.exe (PID 2124) UDP 5353 mDNS	—	windows.netscan

All timestamps are 2019-08-19 process-creation / boot times recovered from the RAM image — they describe when the guest booted and when processes started, not when the intrusion occurred. The memory snapshot itself was taken later that session (DumpIt.exe ran at 14:41:55). Because there is a single image, no intrusion timeline (initial access → exfil) can be anchored; the events below are structural (boot, service stack, injection state, browser sessions) rather than a kill-chain over time.

Agentropix Performance

The full MANUAL triage sequence (case_init → status → evidence_register → pslist → netscan → malfind → cmdline → finding → approval → report_generate) ran end-to-end against the 1.6 GB image over the live MCP. The heavy plugins (malfind, netscan) ran to completion with no timeout. Timings below are the captured wall-clock of each stage from the run; the evidence SHA-256 was computed during registration.

~10

TOTAL RUNTIME

stages (full MANUAL
sequence)

10

MCP TOOL CALLS

MCP tool calls in the
run

1.6 GB

EVIDENCE PROCESSED

1,610,547,200 bytes (raw
RAM)

72/72

RECALL

disk 100% • memory
108/118 (91.5%) –
canonical

STAGE / TOOL	CAPTURE	DURATION	RESULT
case_init	Open case CHALLENGE- NOTCHITUP	—	active, examiner victor.galvan
case_status	Confirm active + indexer	—	active:true, indexer_reachable:true
evidence_register	SHA-256 of 1.6 GB image	—	sha256 80366d7e... 1407b23, 1,610,547,200 B
get_pslist	Process listing + profile auto-detect	—	53 processes, Windows x64 matched
get_netscan	Network sockets	—	97 sockets
get_malfind	RWX VAD / injection hunt	—	4 hits (explorerx2, chrome, WmiPrvSE)
run_volatility (cmd line)	Command lines	—	53 rows (windows.cmdline.CmdL ine)
record_finding	Stage findings	—	5 findings recorded

STAGE / TOOL	CAPTURE	DURATION	RESULT
approve_finding	Examiner gate (simulated)	—	DRAFT → sealed, 5 approved
report_generate	Sealed full report	—	report_id dc580276... 526a, 5 approved findings

Coverage Attestation

This image is a single RAM dump, not a disk corpus, so the disk/memory dataset-recall figures below are the platform's canonical benchmark numbers (regression-suite attestation), not per-case recall for this image. For this case, coverage means: all 53 processes parsed, all 97 sockets parsed, all 4 RWX regions carved+hashed — no plugin timed out or returned partial output.

Disk recall

72/72 (100%)

Memory recall

108/118 (91.5%)

Tests passing

4464

Approval gate

APPROVED-only

Report sealed by the ADR-024 report engine (report_id dc5802763968...) with 5 approved findings, each carrying an HMAC-SHA256 seal. The EvidenceGate enforces APPROVED-only: only examiner-cleared findings reach this report. Chain-of-custody SHA-256 of the source image is recorded and matches across every finding's evidence_refs.

Recommendations

P1 **Isolate notch-it-up and acquire a full disk image.** RAM confirms T1055 injection but cannot show the injector binary, persistence, or initial-access vector — only disk forensics can close those gaps.

P1 **Carve and analyse the explorer.exe RWX payload** (0x4320000, SHA-256 65196e1a...1726ca6f, 64 KB) — disassemble the indirect-jump shellcode and submit the hash to threat-intel for attribution.

P2 **Investigate the WmiPrvSE.exe 512 KB RWX region** (PID 2292, 0x1bd0000) for WMI-based execution/persistence (T1047); review WMI event-consumer subscriptions on the live/disk image.

P2 **Hunt the fleet for the same RWX-VAD injection signature** in explorer.exe / WmiPrvSE.exe across other hosts; deploy a malfind-style RWX VAD detection.

P3 **Confirm the CTF context.** User Jaffa opened `flag.rar` in `pr0t3ct3d\` and ran `Dumplt.exe` — verify whether the injection is the challenge payload vs. a real intrusion before escalating.

Appendix

Methodology

ADR-024 report engine: tool captures → projection → EvidenceGate (APPROVED-only) → sealed sections (executive_summary / findings / iocs / timeline) → HMAC-SHA256 seal per finding. Triage followed the case-activation MANUAL sequence over the live Agentropix-SIFT MCP (Volatility3 wrappers: pslist, netscan, malfind, cmdline). Every datum in this report is sourced from `report_generate(profile=full)` sealed sections or the case EXECUTED-RUN captures; nothing is fabricated. Unprofileable dimensions (on-disk artefacts, dwell time, initial access, C2) are explicitly scoped as not-applicable/inconclusive rather than padded.

Tool & framework versions

Python

3.12+

Volatility 3 Framework

2.28.0

Agentropix-SIFT

71 MCP tools · 16 forensic wrappers

Report engine

ADR-024 (HMAC-SHA256 seal · EvidenceGate APPROVED-only)

mermaid

inlined for offline render

Regression suite

4464 tests

Chain-of-custody hashes

ARTEFACT	ALGO	DIGEST	STATUS
Challenge.raw (source RAM image)	SHA-256	80366d7ec64a5529c95c2f523f4281a5f11efbad33ecb19f73525470c1407b23	✓ registered (chain-of-custody)
explorer.exe 0x4320000 payload (64KB)	SHA-256	65196e1a65d8e4bfcf42f03b7db79cd07a2573f57c6aad40a97c37791726ca6f	✓ carved
explorer.exe 0x3ce0000 payload (4KB)	SHA-256	3da12179ac97a8fdcfd0318ba0da46974e1c1ae648dc37e861a926e8563ca	✓ carved
chrome.exe 0x4830000 payload (4KB)	SHA-256	3243bcc13c7c564288355046c5bc1123888837aeb8ada9ed5cf7800d9c462c1d	✓ carved
WmiPrivSE.exe 0x1bd0000 payload (512KB)	SHA-256	75b4c5d8fc85525f97106ddf84197731942bb9acb2d992de50b8a4b7bcfad9a2	✓ carved
malfind raw stdout	SHA-256	d0a3ae03bf44021de04f7325780226128e94bbea1e6e0decc7866ea3ff2682dd	✓ tool output

Provenance & grounding: all findings carry provenance=MCP and evidence_refs tracing each claim to a Volatility3 plugin output plus the source-image SHA-256. The 21 external IPs were surveyed and classed benign (browser traffic); no fact in this report originates outside this case's real data. MCP host shown as <TAILNET-HOST>; no secrets are included.

Sealed report dc5802763968... · 5 approved findings · examiner victor.galvan · Generated by Agentropix-SIFT report engine (ADR-024) · MCP host · 2026-06-07T03:41:32Z
All data sourced from sealed report_generate(profile=full) sections and the case EXECUTED-RUN captures.
No facts fabricated.